INSIDE THE EARTH

Global earthquake patterns in space, time and depth

DSC5002 Project Team | City University of Hong Kong | Exploratory Data Analysis and Visualization

2000–2025

M ≥ 5 • UTC • Earthquakes only

One frozen catalogue across all panels

46,463
unique earthquake records

70.15%
within 100 km of a PB2002 boundary

83.20%
shallow events (<70 km)

1 INTRODUCTION & METHODS

A global map alone hides variation through time and below the surface. We connect five questions: **where and when, boundary proximity, depth, magnitude frequency, and major-event sequences.**

Data: USGS earthquake-only catalogue [1], 1 Jan 2000 to 31 Dec 2025; preferred M ≥ 5. Event IDs are unique; four negative reported depths are retained. PB2002 [2] supplies 241 boundary features; Natural Earth [3] supplies map context.

Methods: shortest spherical distance to boundary arcs (R = 6,371.0088 km); six geographic boxes plus Other; full-sample statistics; equal 13-year comparisons adjusted for 4,749 versus 4,748 days. No prediction model.

2 WHERE ARE THE EVENTS? · RQ1–2

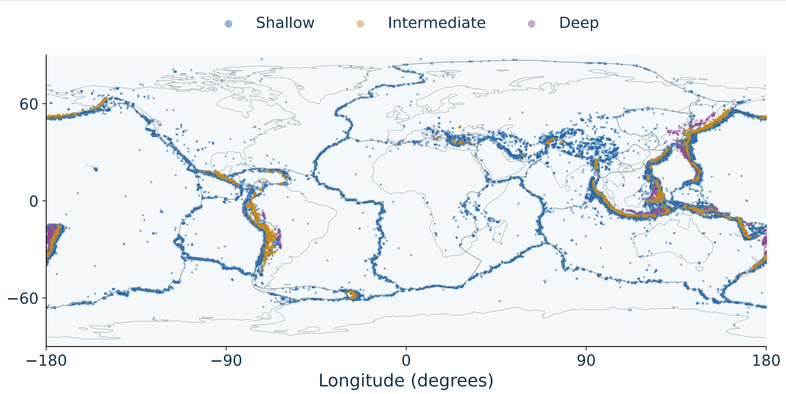


Figure 1. All 46,463 epicentres; points coloured by depth. Grey lines show PB2002 boundaries. The circum-Pacific pattern is prominent. Overplotting indicates concentration, not an area-normalized risk estimate.

3 HOW CLOSE TO BOUNDARIES? · RQ2

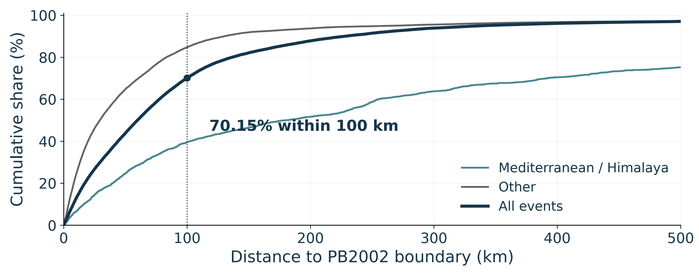


Figure 2. Median distance = **59.68 km**; 70.15% lie within 100 km. These are distances to a simplified, dated map layer, not to every active fault. No spatial null model was fitted.

4 RECORDED ACTIVITY OVER TIME · RQ1

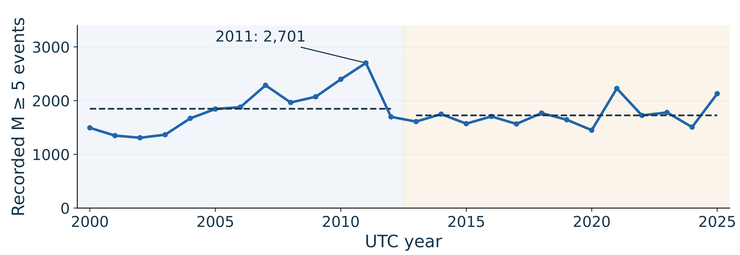


Figure 3. Annual M ≥ 5 counts; dashed lines are period means. The maximum is 2,701 in 2011. Later-period recorded rate: **−6.63% at M ≥ 5**, **−15.42% at M ≥ 6**. These are descriptive comparisons, not a monotonic trend.

5 REGIONAL CHANGES · RQ1

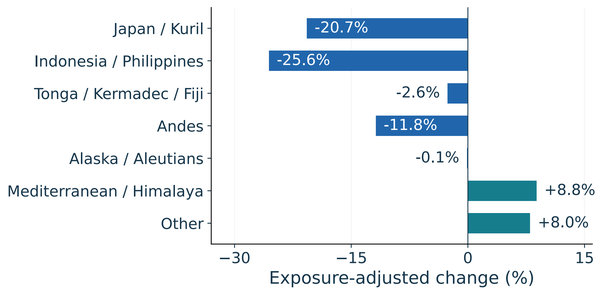


Figure 4. 2013–2025 versus 2000–2012, M ≥ 5; equal exposure rates. Indonesia/Philippines decreases 25.58%; Mediterranean/Himalaya increases 8.82%. Boxes differ in area and do not define tectonic plates.

6 DEPTH DIFFERS BY REGION · RQ3

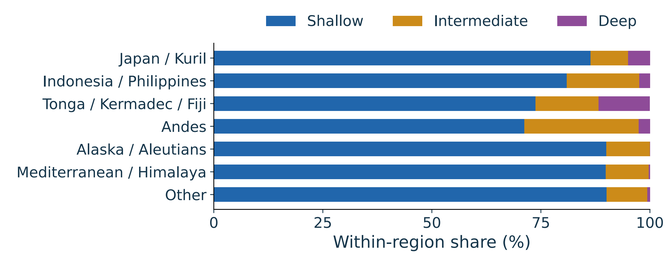


Figure 5. Shallow <70, intermediate 70–<300, deep ≥300 km. Tonga/Kermadec/Fiji has **11.76% deep** events versus Japan/Kuril 5.01%; Andes has 26.19% intermediate. Each row uses its own known-depth denominator.

7 MAGNITUDE & FREQUENCY · RQ4

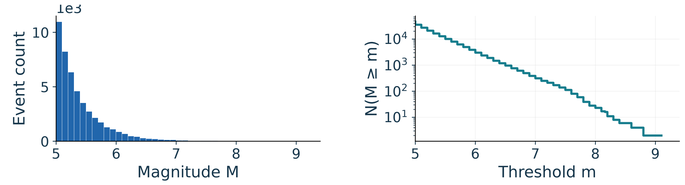


Figure 6. Histogram and inclusive cumulative counts (log axis): M ≥ 6: **3,874**; M ≥ 7: **387**; M ≥ 8: **28**. Mixed magnitude types are not homogenized; M5 completeness is not established.

8 LOCAL DEPTH SECTIONS · RQ3

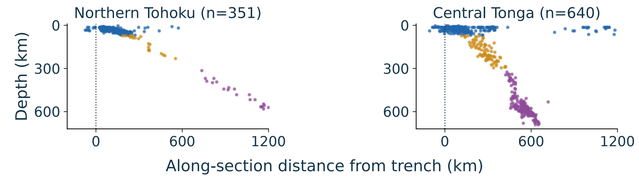


Figure 7. ±100 km finite swaths; zero at the mapped trench. Equal kilometre scales, depth positive downward. Deep events: Tohoku **21/351**, Tonga **186/640**. Routes were explored and selected; profiles are illustrative.

9 MAJOR-EVENT SEQUENCES · RQ5

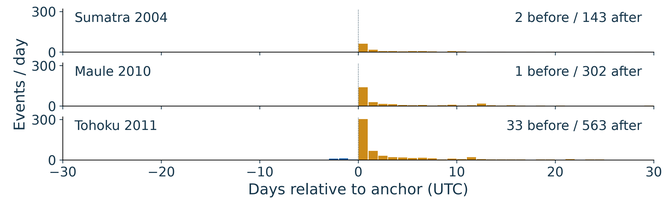


Figure 8. Fixed 500 km, ±30-day windows; anchors excluded. First 3 post-event days contain **60.1%, 60.9%, 70.9%** of post-window counts. Shared y-scale; blue = before, gold = after.

PLAY THE ANIMATION > (click here or the image)

VictoriaSU551.github.io/DSC5002-Earthquake-Explorer/

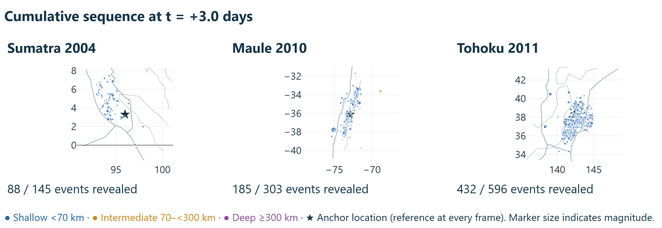


Figure 9. Actual animation keyframe at +3 days, cumulative mode. Full interactive companion includes play, time slider and 7-day trailing windows.

CONCLUSION & LIMITS Boundary proximity and depth structure are clear catalogue patterns. Cross-period counts need caution: Mw-family share changes from 73.45% to 55.05%, and reporting fields change. These results do not establish causality, completeness or future hazard.

[1] USGS FDSN Event Web Service: earthquake.usgs.gov/fdsnws/event/1/ · [2] Bird (2003), doi:10.1029/2001GC000252; PB2002 GeoJSON, ODC-By · [3] Natural Earth 110m, public domain.